Managed Global Analytics Solution

Design Document

Table of Contents

Team Roles

Overview

- Problem Definition
- Solution Requirements
- Out of Scope
- Assumptions

Solution

- Security Considerations
- Cost Analysis
- Cross-region Considerations
- Operational Readiness Considerations
- Work Required
- High-Level Test Plan
- Milestones

Data

Glossary

Team Roles

Riley Hopkins
Project Manager, Pipeliner
Cecilia Cabiya
MicroStrategy Visualizations
Joeny Vo
Data Lake Management, Pipeliner
Benjamin Kalnbach
Lead Pipeliner

Project Workflow Management on our [Trello Board]

Overview

This document outlines the technical aspects and scope of a solution being presented to the client, a company in the Big Data space, who looks to migrate their system to a cloud-based infrastructure. The company focuses on economic indicators in various countries, as well as requiring that non-economic, health related indicators can be stored and analyzed alongside the economic indicators. As a result, the client's primary focus is to manage vast amounts of data, analyze it, and visualize the analysis in reports to be presented to policymakers – hence the client requires strong analysis, storage capacity, and visualization systems.

To support these needs, Team Juniper has developed a solution to migrate the company's data into a cloud-based infrastructure. To house, analyze, and visualize the client's health and wellness data indicators, Team Juniper has developed a pipeline and visualization process the client can utilize, using technologies including Azure Data Lake Storage, Alteryx Designer Cloud, MicroStrategy ONE, and Databricks. Geographic visualization of the health-related indicators is also supported by Team Juniper's solution, allowing a broader or narrower scope for reports.

Problem Definition

The client's company gathers large amounts of data to produce reports, but is looking to move their on-premises data storage into the cloud. In addition to the storage itself, the data pipelining process is to be cloud-based, as well as new methods of forming visualizations needed for the company's reports.

On premises infrastructures can have serious drawbacks, which the client is looking to avoid. There can be additional overhead costs, added management tasks, and more physical space than

a company is willing, so Team Juniper's cloud-based solution will aim to eliminate these on-premises issues.

Solution Requirements

In migrating a large-scale company infrastructure, several considerations for the data's flow, storage, security, and analysis-preparedness have to be made. In this section, requirements that the solution has to address have been laid out, to illustrate where Team Juniper's solution focuses their solution.

Requirement One: Data Pipelining
In order to get data from the client's various sources and consolidate it into one storage system, a pipeline must be crafted that moves and transforms data. The pipeline must be at least partially automated, such that data transference does not need to be watched in order to succeed. In addition, connections made by this pipeline must be secure, using HTTPS or another secure transfer protocol.

Requirement Two: Storage
Any cloud storage system proposed must have equal or greater capacity to the on-premises storage system currently employed by the client. In addition, to solve the client's heavy management necessities, any storage solution will have to abstract some management tasks to the cloud host, while still allowing any desired changes to be made.

Requirement Three: Visualization
The client's role as an advisor for various policy makers necessitates that visualization tools are made available to data analysts to present with. Any solution being utilized by the client will need a visualization tool as well. This tool will need the ability to intake large amounts of data and provide a variety of graph options to illustrate different points. The dataset being worked with is global economic and health data, and any visualization tools must also be able to geographically display this data with a map or geography tool.

Out of Scope

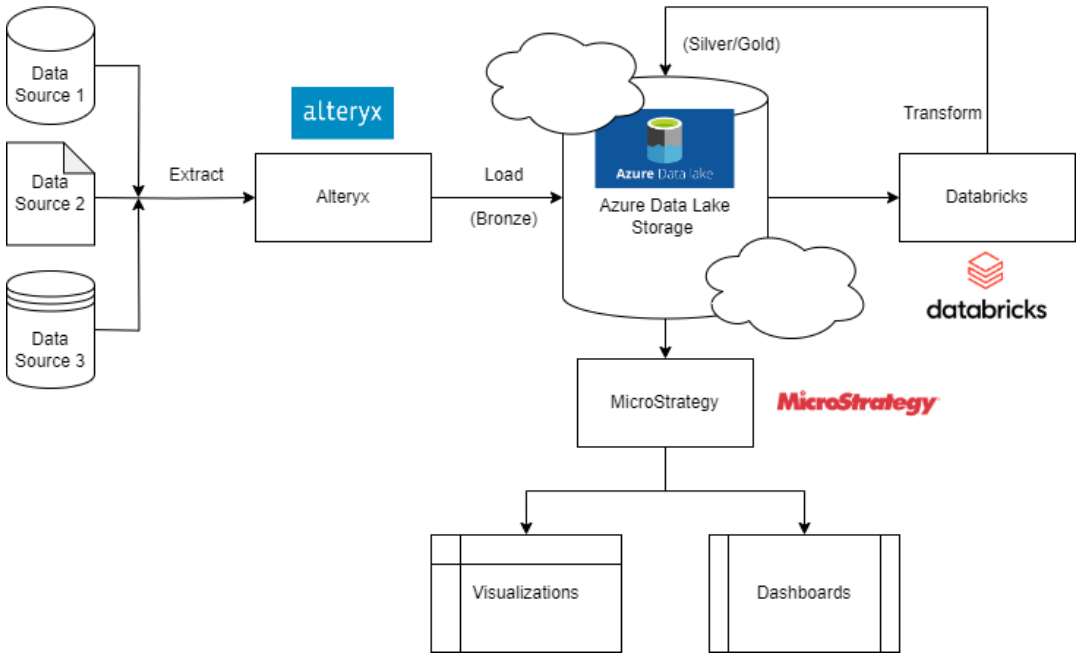
The scope of this solution is limited to processing and visualizing data. This solution is not intended to propose how to gather the data obtained from API/Data Sources. While the end user is intended to be able to use the data and visualizations to inform their decisions with respect to legislation, this proposal is not in of itself designed to make such decisions. Additionally, the indicators and metrics that are part of this solution are intended to give a general picture of a given country's economic development, but by no means encompass all aspects pertaining to a country's economic development.

Assumptions

Solution assumes several services are available for use upon deployment. These services include: Azure Databricks, ADLSv2, Alteryx Designer Cloud and MicroStrategy ONE. Should any one of these services be down, the proposed solution should be assumed non-operational or limited to certain operations. Additionally it is assumed that source API/Data Sources are operational, if no longer operational, the solution will be unable to update current data beyond its current state. This particular issue may be remedied through alterations to the pipeline to provide new, updated, data sources from which to extract up to date data.

Beyond external services the proposed solution also assumes, at the very minimum, that the monetary cost and personnel estimates are met. These estimations and a high level breakdown of each can be found in the Cost Analysis and Work Required sections, respectively, within this document.

Solution



At a surface level the proposed solution will deliver interactive geospatial visualizations allowing users to select specific countries via MicroStrategy ONE. The solution will also deliver several elements that enable the user visible aspects of this solution. These include an entity relationship diagram, managed Azure data lake, Alteryx Cloud ETL Pipelines, and Databrick Spark Notebooks.

Looking closer at the solution, Alteryx Pipelines is our starting point in our overall solution. Alteryx Cloud Pipelines will enable the initial raw and unorganized data to be read into our Data Lake's Landing Zone. From our Data Lake's Landing Zone we will continue to use Alteryx to pipeline our data into the Bronze Layer, where our data is sorted into a hierarchical namespace. The solution demo will use API's/Data Sets from publically available online sources, however the pipeline can easily be adapted to use locally stored data and read it into the Data Lake's Landing Zone.

Once in the Bronze Layer, Databricks will be used to do the remaining transitions from our Bronze to Silver Layer, and from our Silver to Gold Layer. The Bronze to Silver process will involve removing superfluous columns for data, combining data that comes in several parts into a single entity, and renaming our columns so that the data's meaning is more readily apparent. The Silver to Gold process will be our most time and resource intensive task within Databricks. This will require creating a denormalized schema with dimension table consisting of the relevant data needed for analysis/aggregations and a central fact table containing all of our aggregations needed to create informative visualizations.

Once our data has successfully been transformed into our Gold Layer, the resulting data will be loaded into MicroStrategy ONE to be used to create our interactive geospatial visualization and dashboards. The user will then be able to use this visualization to view key economic metrics, indicators, and visualizations for a selected country for a given year. Additionally, non-economic indicators and metrics will also be visible in order to inform the user of how non-economic factors can impact a country's economy.

The demo of this solution will include each of the steps above, but will not demonstrate one key factor to a successful long-term deployment of said solution. This solution requires individual(s) who are able to continue to maintain the Data Lake and Compute Cluster post deployment. Additionally, while the processes within this solution are intended to be automated, they do not ensure the solution will be future proof. A smaller team of individuals able to address errors that occur as the result of unintended operations or data with changed formatting, is necessary to ensure the future success of this solution.

Security Considerations

All data in the Data Lake will be encrypted both in transit and at rest, as per the specifications of the client. Azure's built-in encryption services will be used for encrypting the data and all data will only be accessed through these encrypted protocols. Any APIs will use secure HTTPS protocol. All services will be closely managed by the team to ensure only the client and team members have access to data, pipelines, and dashboards.

Cost Analysis

The primary cost will be the cost of running the Databricks Cluster for the pipeline. Another consideration is the cost of running the ADLS Gen 2 data lake. This will likely be much lower but may vary, as it is dependent on the size of the data sources.

Estimated Cost Breakdown:
ADLS Data Lake cost: ~$0.93 monthly
Databricks Cluster cost: ~$319.83 monthly
Total: ~$320.76 monthly

Cross-region Considerations

All services will be hosted in the cloud with Azure, and can be accessed from any region. This ensures maximum accessibility to the client, and speedy redeployment if new requirements are made.

Operational Readiness Considerations

The solution will be implemented in Azure using the best practices. Utilizing Azure's infrastructure means an agile development cycle which can be continuously deployed and iterated upon. It also ensures the integrity and security of all data within the pipeline.

Work Required

- Azure Data Lake
  - Create an Azure Storage Account for ADLS
  - Set up a Medallion Architecture system using our container
- Data Collection and Integration
  - Gather data from sources
  - Transfer raw data to landing zone in ADLS
- Data Pipeline
  - Develop workflows in Alteryx for simple transformations
    - Landing Zone → Bronze
  - Create Databricks notebooks to further transform our data with more complex joins, cleaning, and filtering
    - Bronze → Silver
  - Aggregate and transform data into star schema for OLAP
    - Silver → Gold
- Data Analysis and Visualization

    - Extract meaningful insights from our data
    - Design interactive geospatial dashboards using MicroStrategy
    - Features for drilling up and down into a specific country data
    - Customizable by year
    - Include our main economic metrics as well as all our other additional metrics

High-Level Test Plan

- Azure Data Lake
    - Ensure our medallion layers are correctly implemented
    - Validate data transformations between layers
- Data Collection and Integration
    - Verify that data is complete for each data source
    - Ensure files are in correct format (Parquet)
- Data Pipeline
    - Ensure data quality by cleaning, transforming, filtering, etc
    - Test Alteryx workflow for loading, transforming, and writing
    - Load testing to ensure an efficient Spark configuration
    - Error handling
- Data Analysis and Visualization
    - Ensure dashboards are accurate
    - Test interactive features like drilling up and down
    - Test geospatial visualizations for each country

Milestones

1. Project Planning
    a. Determine roles/responsibilities for each team member
    b. Set up a project management tool (Trello)
    c. Find additional non-economic metrics and data sources
    d. Complete Design Document
2. Data Lake Architecture Setup
    a. Create an Azure Storage Account and blob container
    b. Ensure correct storage account configuration
        i. Encrypted at rest and in transit
        ii. Only accessible through secure methods
    c. Set up medallion layers in Data Lake
    d. Create directories for our data sources
3. Data Collection and Integration
    a. Gather data from all sources
    b. Store raw data in Landing Zone on our Data Lake

4. Data Pipeline
    a. Design workflows in Alteryx to extract transform and load data
    b. Implement data quality and cleaning procedures
    c. Create Databricks notebooks to handle complex joins and aggregations
    d. Transform data into star schema for analysis
5. Data Analysis and Visualization
    a. Extract insights from our metrics
    b. Design geospatial visualizations and dashboards on MicroStrategy
    c. Implement drilling down functionality for visualization
    d. Create an Entity Relationship Diagram (ERD) describing how our data and metrics are connected
6. Validation
    a. Test entire data pipeline from data ingestion to visualization
    b. Ensure data is accurate throughout pipeline
    c. Confirm all project requirements are met

Data

Below is a list of data sources, including the metrics and datasets that are pulled from each source.

- World Bank – World Development Indicators (WDI)
    - Metrics
        - Gross Domestic Product (GDP)
            - Total value of goods produced
        - Gross National Income (GNI)
            - Total income gained by country's residents
        - Consumer price index - inflation
        - Gini index - income inequality
        - Unemployment / labor force participation rate
        - Poverty Ratio
        - Government expenditure

- Cause of Deaths around the World - Kaggle
- Our World in Data, Energy
    - Metrics
        - Energy Sources
            - Energy source output by country and year
        - Energy Access
            - Percentage of population with energy access by country and year

        - Other links
            - Github
- ILO - International Labour Organization
    - Metrics
        - Fatal Injuries
            - Fatal injuries by sex and migrant status
            - Fatal injuries by industry
        - Working Time
            - Working time by industry, age, and sex
- Food and Agriculture Organization (FAO)
    - Metrics
        - Food Security Indicators
            - Prevalence of undernourishment by country and year
            - Severe food insecurity percentage by country and year
        - Food Prices Index
            - By country, year, and month

Glossary

This glossary includes any technologies frequently mentioned in the document, commonly used acronyms, and other terms that may be useful to refer back to.

ADLS Gen 2 – Azure Data Lake Storage Gen 2: Azure's enterprise-level data lake solution, in which the client's data and files will be stored as BLOBs.
Alteryx: A data analytics platform, of which Team Juniper will be utilizing their pipelining services. Alteryx's pipelining tool is a visual way to craft data flows, and allow for input and output connection points to automate the process of reading, cleaning, and delivering data to its destination.
BLOB – Binary Large Object: An object stored in binary, allowing any type of file to be stored, and comes in various sizes, acting as a flexible and versatile file storage method. BLOBs are utilized by Azure's data lakes to ensure many kinds of files can be safely and reliably stored.
Data Lake: A storage infrastructure in which a variety of different data types can be stored, from unstructured image files to structured tabular files. Data lakes are where an enterprise's data is put for long term storage.
Medallion Architecture: A way of describing file readability and preparedness for analysis; how "cleaned" a file is. Data is stored in different medallion levels, and as a file is further prepared and standardized, a copy of the file is moved further in the medallion architecture, signifying the file's further preparation. The

medallion architecture is composed of four levels: the Landing Zone, Bronze Layer, Silver Layer, and Gold Layer.

- ***Landing Zone:*** Where data and files loaded in directly from the pipeline are housed, without any specific format or organization.
- ***Bronze Layer:*** Where data and files that have been moderately organized to fit the overall structure of the data lake where they are housed. There is no specific file format used at this layer, nor any major transformations or modifications to the data.
- ***Silver Layer:*** Where data that has been cleaned in a processing tool is housed. At this layer, all files should be in the parquet file format, and the processing tool used to get data to this stage for this project is Databricks.
- ***Gold Layer:*** Where data that is fully prepared for visualization and analysis is housed, modified to fit a star schema and remaining in the parquet file format.

Microstrategy ONE: A visualization tool into which OLAP-ready data can be input, and Data Analysts can visualize and analyze the client's data.

OLAP – Online Analytical Processing: The process by which read-optimized and highly structured data is analyzed by Data Analysts.